Phys 153: Fundamentals of Physics III

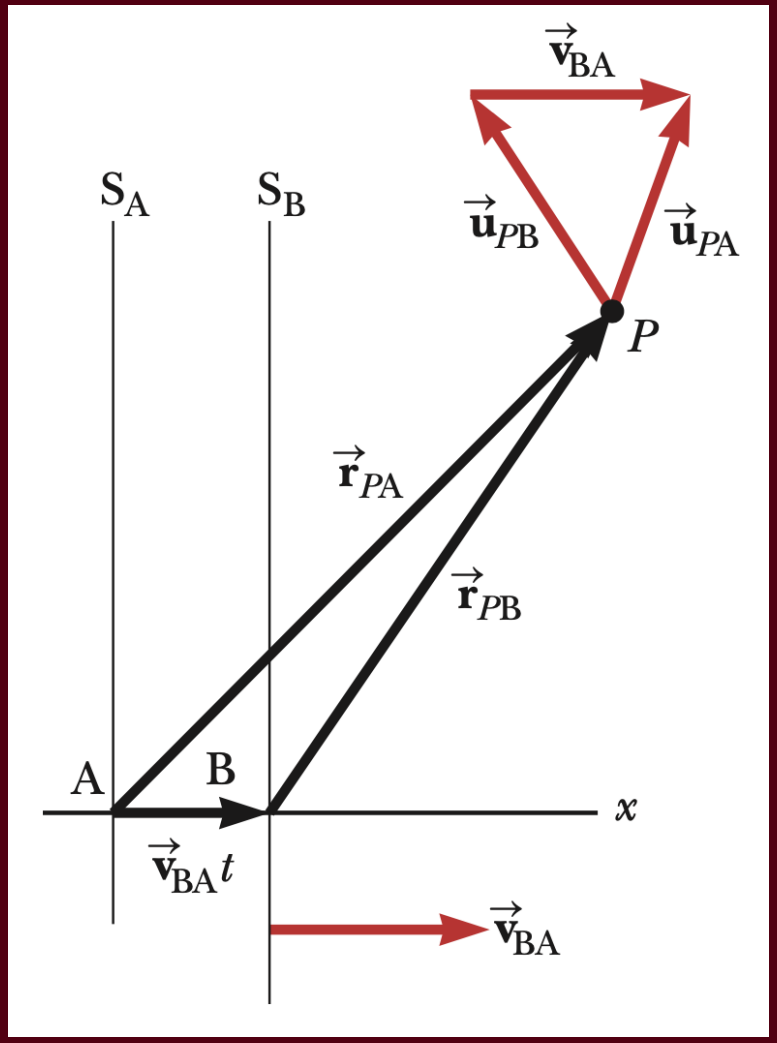
Unit #5 – Velocity Addition & Relativistic Dynamics

Aaron Wirthwein

Spring 2027

Galilean Velocity Addition

Recall from PHYS 151 (Chapter 4 of Serway & Jewett)



$$\vec{\mathbf{r}}_{PA} = \vec{\mathbf{r}}_{PB} + \vec{\mathbf{v}}_{BA}t$$

$$\vec{\mathbf{u}}_{PA} = \vec{\mathbf{u}}_{PB} + \vec{\mathbf{v}}_{BA}$$

Relativistic Velocity Addition

Particle moves with x-velocity u'_x in S'. The velocity of the same particle in S can be found with the inverse transformation:

$$u_x \equiv \frac{dx}{dt} = \frac{\gamma(dx' + vdt')}{\gamma(dt' + vdx'/c^2)}$$

$$u_x = \frac{u'_x + v}{1 + vu'_x/c^2}$$

$$u_y = \frac{u'_y}{\gamma(1 + vu'_x/c^2)}$$

$$u_z = \frac{u'_z}{\gamma(1 + vu'_x/c^2)}$$

Velocity Addition by Composition

Consider the composition of two transformations, one from the Home Frame to the Other Frame, and another transformation from the Other Frame to the rest frame of a particle moving only along the x axis:

$$\gamma_3 \begin{pmatrix} 1 & -\beta_3 \\ -\beta_3 & 1 \end{pmatrix} = \gamma_2 \begin{pmatrix} 1 & -\beta_2 \\ -\beta_2 & 1 \end{pmatrix} \gamma_1 \begin{pmatrix} 1 & -\beta_1 \\ -\beta_1 & 1 \end{pmatrix}$$

$$= \gamma_1\gamma_2 \begin{pmatrix} 1+\beta_1\beta_2 & -\beta_1-\beta_2 \\ -\beta_1-\beta_2 & 1+\beta_1\beta_2 \end{pmatrix}$$

$$\gamma_3\beta_3 = \gamma_1\gamma_2(\beta_1+\beta_2)$$

$$\gamma_3 = \gamma_1\gamma_2(1+\beta_1\beta_2)$$

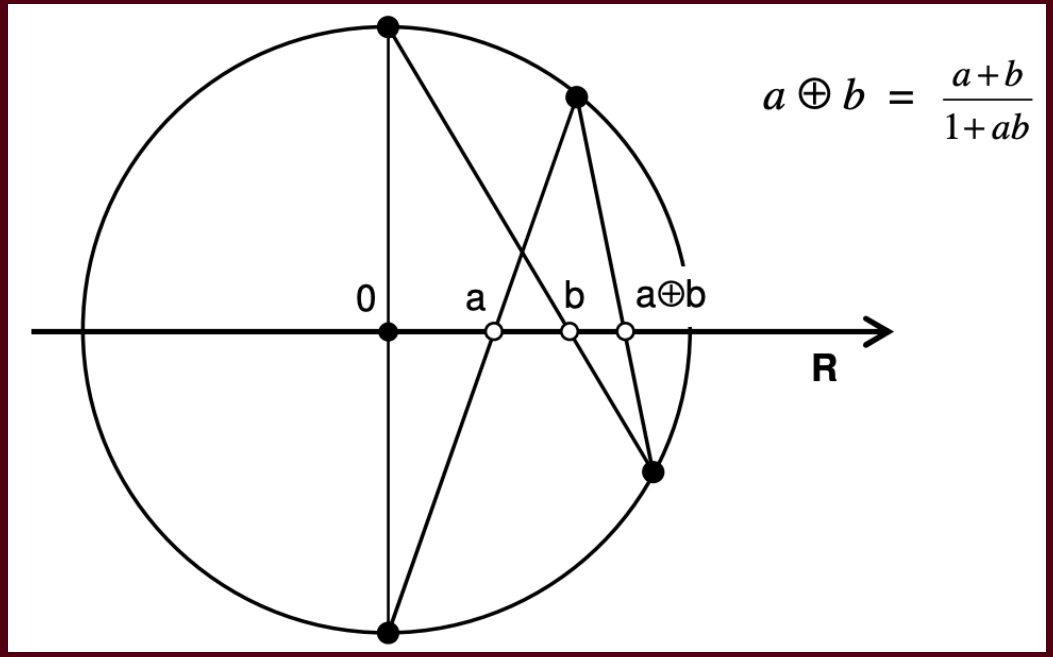
$$\beta_3 = \frac{\beta_1+\beta_2}{1+\beta_1\beta_2}$$

Translate β's into u's and v's (last slide)

$$u_x = \frac{u'_x + v}{1 + vu'_x/c^2}$$

This works even if the particle's velocity changes—just consider an instantaneous rest frame for each moment along the particle's worldline

Geometric Velocity Addition

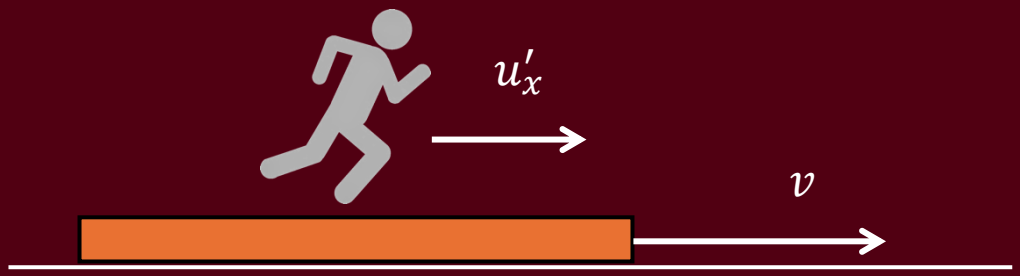


$$\frac{\frac{1}{2}+\frac{1}{2}}{1+\frac{1}{2}\cdot\frac{1}{2}} = \frac{4}{5} < 1$$

$$\frac{\frac{3}{4}+\frac{1}{2}}{1+\frac{3}{4}\cdot\frac{1}{2}} = \frac{10}{11} < 1$$

$$\frac{1+\frac{1}{2}}{1+1\cdot\frac{1}{2}} = 1$$

$$\frac{1+1}{1+1\cdot 1} = 1$$



Henri Poincaré (1854 – 1912)

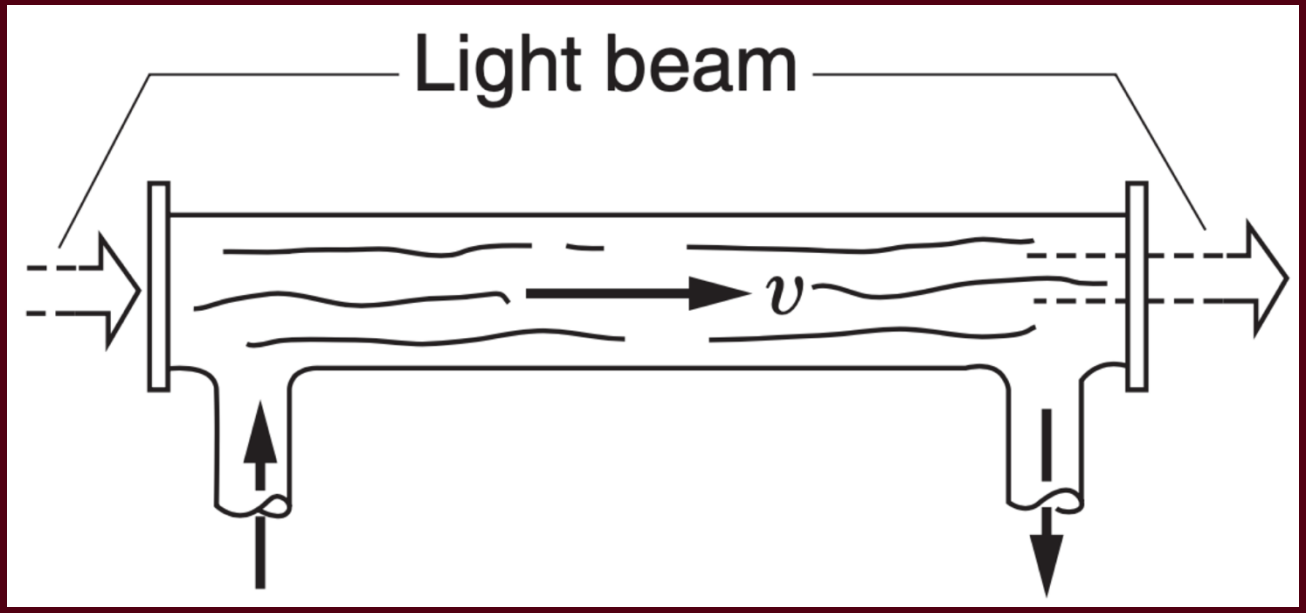
Prominent French physicist and mathematician who made significant contributions to relativity:

- Viewed time dilation and length contraction as dynamical effects due to motion through ether
- May have been the first to formulate the principal of relativity, discover the invariant interval s^2, and introduce 'four vectors'
- Brought the mathematics of group theory to physics and recognized that Lorentz transformations form a group
- Derived relativistic velocity addition



Example

Consider a tube filled with water. If the water is at rest, the speed of light in the water with respect to the laboratory is $u = c/n$. What is the speed of light when the water is flowing with speed v?



Four Velocity

It's annoying that ordinary velocity doesn't transform the same way coordinates do. On the other hand, we'll find that a new quantity called four velocity does.

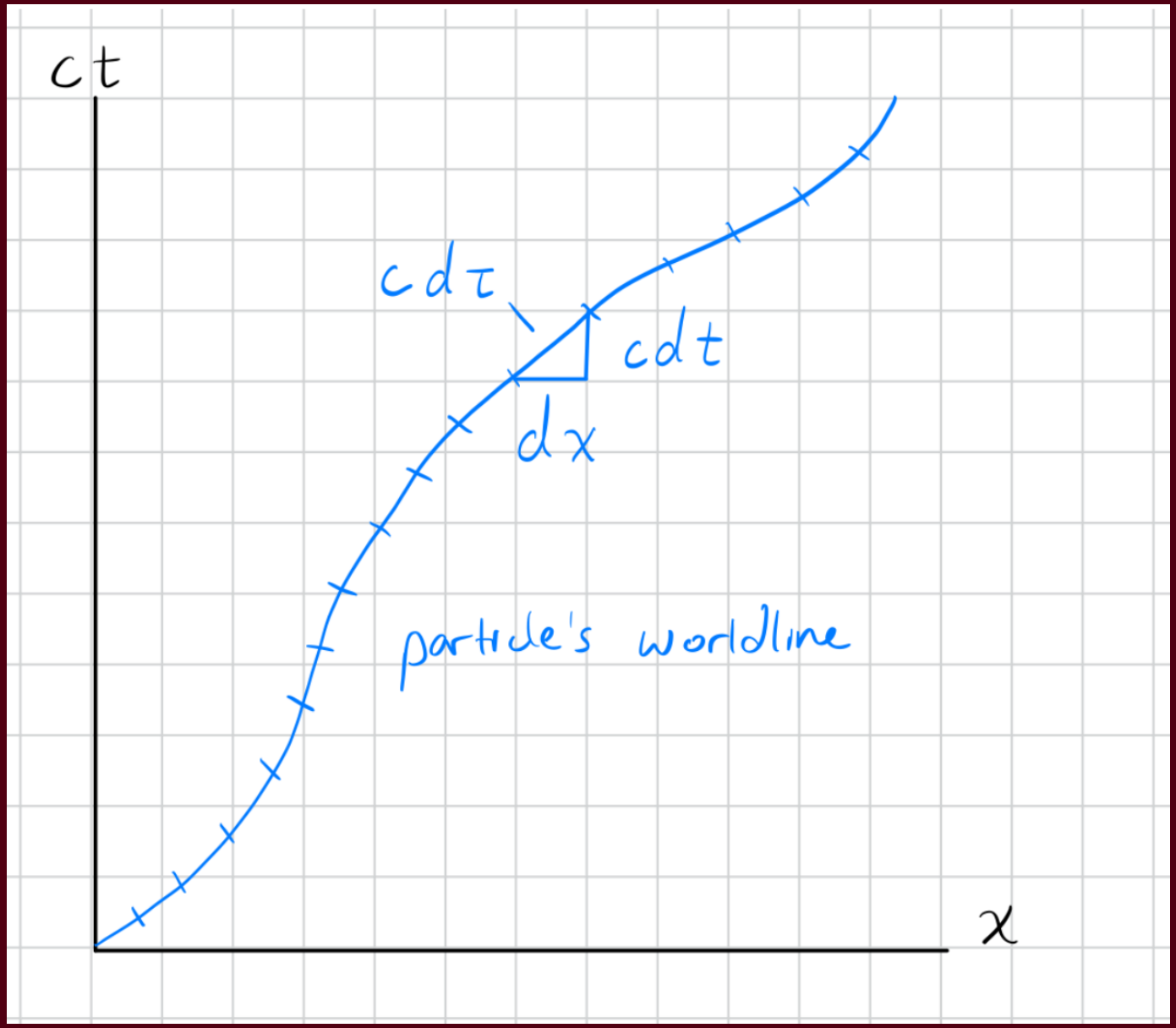
Consider an infinitesimal spacetime interval between two successive events in a massive particle's worldline:

$$0 < ds^2 = c^2 d\tau^2 = c^2 dt^2 - dx^2 - dy^2 - dz^2$$

Proper time interval

$$c^2 = c^2 \left(\frac{dt}{d\tau}\right)^2 - \left(\frac{dx}{d\tau}\right)^2 - \left(\frac{dy}{d\tau}\right)^2 - \left(\frac{dz}{d\tau}\right)^2$$

Components of the four velocity


ct
cdτ
cdt
dx
particle's worldline
x

Four Velocity

The **four velocity** of a particle is

$$\mathbf{U} \equiv \gamma(u)(u_x, u_y, u_z, c)$$

$$\frac{dt}{d\tau} = \gamma(u) \qquad \frac{dx}{d\tau} = \frac{dx}{dt} \cdot \frac{dt}{d\tau} = \gamma(u) u_x$$

$$\begin{pmatrix} U_x' \\ U_t' \end{pmatrix} = \gamma(v) \begin{pmatrix} 1 & -v/c \\ -v/c & 1 \end{pmatrix} \begin{pmatrix} U_x \\ U_t \end{pmatrix}$$

The four velocity components transform like spacetime coordinates! Generally, a four vector is any 4-dimensional vector that transforms between frames according to the Lorentz transformation. There are many examples.

Invariant Norm

Different inertial observers generally assign different components to the four velocity.

Nevertheless, they agree on its invariant norm, which is analogous to the magnitude of a three vector:

$$\mathbf{U} \cdot \mathbf{U} = c^2 \left(\frac{dt}{d\tau}\right)^2 - \left(\frac{dx}{d\tau}\right)^2 - \left(\frac{dy}{d\tau}\right)^2 - \left(\frac{dz}{d\tau}\right)^2 = \boldsymbol{c^2}$$

The norm-squared of any four vector is a Lorentz-invariant quantity. Although the components of physical quantities depend on the observer, Lorentz-invariant quantities have the same value for all observers.

Principle of Covariance

- The principle of relativity states that the laws of physics are the same for all inertial observers; no inertial frame is physically preferred.
- The principle of covariance expresses this mathematically; when physical quantities are transformed from one inertial frame to another, the equations describing the laws of physics must retain the same form.
- Thus, the principle of relativity is the physical statement, while Lorentz covariance is its mathematical implementation in special relativity.

Four Momentum

Ordinary Newtonian momentum $m\vec{\mathbf{u}}$ does not obey a conservation law that is valid in all inertial frames!

A particle of mass m has a four momentum $\mathbf{P} = m\mathbf{U}$ which is conserved in the absence of forces, and similarly, the total four momentum of an isolated system of particles is conserved. The conservation of four momentum is therefore a frame-independent law of physics.

Invariant norm of four momentum follows directly from invariant norm of four velocity (multiply by m^2):

$$m^2c^2 = c^2\left(m\frac{dt}{d\tau}\right)^2 - \left(m\frac{dx}{d\tau}\right)^2 - \left(m\frac{dy}{d\tau}\right)^2 - \left(m\frac{dz}{d\tau}\right)^2$$

$$m^2c^2 = (\gamma mc)^2 - (\gamma m\vec{\mathbf{u}})^2$$

Example

An object with a mass of 1.0 kg moves with x-velocity $u_x = \frac{4}{5}c$ in the Home Frame. What are the components of its four momentum in this frame? What are the components of its four momentum in the frame where it's at rest?

Relativistic Momentum

$$m^2c^2 = (\gamma mc)^2 - (\gamma m\vec{\mathbf{u}})^2$$

The spatial components of $\mathbf{P}$ are related to Newtonian momentum. A particle's **relativistic momentum** is

$$\vec{\mathbf{p}} = \gamma\, m\, \vec{\mathbf{u}}$$

The total relativistic momentum of an isolated system is conserved (follows from conservation of four momentum).

The relativistic momentum reduces to Newtonian momentum in the non-relativistic limit (go figure).

Conceptual Question

When an object of mass m is pushed to relativistic speed u, its momentum is

A. greater than mu.

B. smaller than mu.

C. equal to mu.

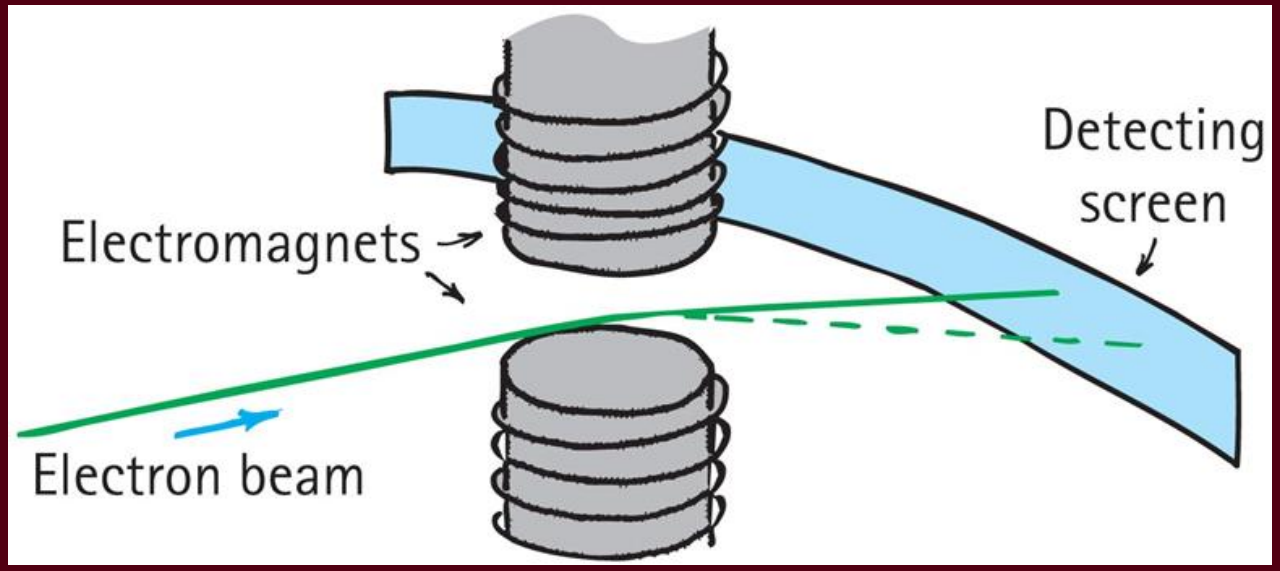
Conceptual Question

When an object of mass m is pushed to relativistic speed u, its momentum is

A. greater than mu.

B. smaller than mu.

C. equal to mu.



Relativistic Energy

$$m^2c^2 = (\gamma mc)^2 - (\gamma m\vec{\mathbf{u}})^2$$

What about the time component of the four momentum, the quantity γmc? Let's consider $u \ll c$ (low-speed limit)

$$\gamma mc = mc\left(1 - \frac{u^2}{c^2}\right)^{-1/2} \approx mc + \frac{1}{c}\left(\frac{1}{2}mu^2\right) + \cdots$$

Kinetic energy!

Define relativistic energy as

$$E = \gamma mc^2 = \frac{mc^2}{\sqrt{1 - u^2/c^2}}$$

Kinetic Energy & Rest Energy

The equation for relativistic energy implies that even when the particle isn't moving, it has rest energy mc^2 (finally!)

Define relativistic kinetic energy as

$$K = E - mc^2 = (\gamma - 1)mc^2$$

When $u \ll c$, we find Newtonian kinetic energy $K = \frac{1}{2}mu^2$

Energy-Momentum Relation

From the invariant norm of four momentum (which can be traced back to the metric equation), substitute expressions for relativistic energy and momentum:

$$m^2c^2 = (\gamma mc)^2 - (\gamma m\vec{\mathbf{u}})^2$$

$$m^2c^2 = (E/c)^2 - p^2$$

$$\boxed{E^2 = (pc)^2 + (mc^2)^2}$$

$$p^2 = \vec{\mathbf{p}} \cdot \vec{\mathbf{p}} = p_x^2 + p_y^2 + p_z^2$$

Example

(a) Use the energy-momentum relation to show that $pc = \sqrt{K(K + 2mc^2)}$, where K is the relativistic kinetic energy.

(b) Show that for a massive particle moving slowly compared to the speed of light, K/mc^2 is a very small quantity (second order in v/c).

(c) When $K \ll mc^2$, show that $p = \sqrt{2mK}$, consistent with Newtonian physics.

A Useful Trick

We use the conservation principle to find energy and momentum, and once we have those, we can find the velocity components. Energy and momentum are highly nonlinear expressions in the velocity components, but we can use the following trick to find them:

$$\frac{p_x c}{E} = \frac{\gamma m u_x c}{\gamma m c^2} = \frac{u_x}{c}$$

$$u_x = \frac{p_x c^2}{E} \qquad u_y = \frac{p_y c^2}{E} \qquad u_z = \frac{p_z c^2}{E}$$

$$u = \frac{p c^2}{E}$$

In terms of magnitudes u and p

Energy-Momentum for Light

Consider a ‘photon’ of light traveling along some trajectory:

$$c = \frac{pc^2}{E} \Rightarrow \boxed{E = pc}$$

$$E^2 = (pc)^2 = (pc)^2 + (mc^2)^2$$

$\boxed{m = 0}$ Light has no mass!

$p_x = \pm E/c$ 1D motion of a light pulse

Consequences for Dynamics

- The postulates of special relativity call for revisions to core concepts of Newtonian physics.
- Force, momentum, and energy are all redefined so that if energy and momentum are conserved in one inertial frame of reference, they are conserved in all inertial frames of reference. (principle of covariance!)
- Just as space and time make up a single concept of spacetime, energy and momentum make up a single quantity composed of *relativistic energy* and *relativistic momentum (four momentum)*.
- The mass of a single particle is a relativistic invariant, but the total mass of a system of particles is not. The total mass of a system of particles is frame-dependent!

Conceptual Question

Two observers are in uniform relative motion. Which of the following quantities are **frame independent** (both observers measure the same value)? Select all the apply.

A. Velocity of a single particle
B. Mass of a single particle
C. Total energy of a system of particles
D. Total four momentum of a system of particles
E. Total mass of a system of particles

Example

Somewhere in deep space, an indestructible crystal with mass $m_1 = 12$ kg is moving in the $+x$ direction with $u_{1x} = +4c/5$ in some inertial frame. The crystal strikes another crystal of mass $m_2 = 28$ kg at rest. The crystals elastically bounce off each other and the less massive one moves with velocity $u'_{1x} = -5c/13$.

What is the x-velocity of the larger crystal after the collision? Show that the collision does not conserve Newtonian momentum.

Example

Two balls of putty, each of mass m, move in opposite directions toward each other with speeds of $0.95c$. The balls stick together, forming a single motionless ball of putty at rest.

What is the mass of this ball of putty? Express your answer as a multiple of m.